Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [14EE61]

SIXTH SEMESTER B. Tech UNIVERSITY EXAMINATION, JUNE–2018

POWER SYSTEM ANALYSIS – I

Time:3 Hours **Maximum Marks:70**

Instructions: 1. Answer any five full questions by selecting at least two questions from each part.
2. Any missing data may be suitably assumed.

PART - A

1. a. Explain the significance of single line diagram of Power systems and discuss the various standard symbols used in it. **04**

b. Compare a three winding transformer with a two winding transformer and write the similarities and dissimilarities between them. **02**

c. The single line diagram of an unloaded power system is shown in fig. reactances of the two sections of the transmission line are shown on the diagram. The generators and the transformers are rated as follows: Draw the pu reactance diagram on a base of 50 MVA, 13.8 kV. **08**

Generator !: 20 MVA, 13.8kV, Xd"=0.20 pu.

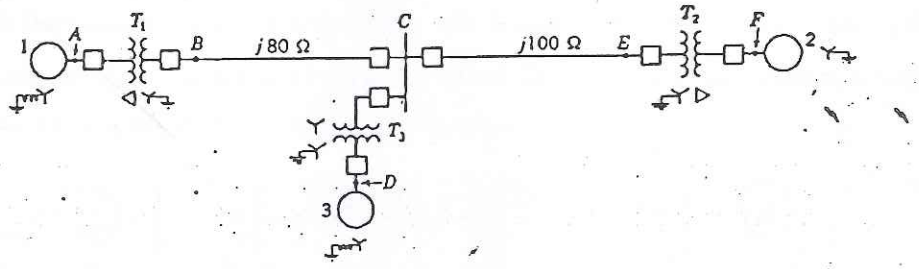
Generator 2: 30 MVA, 18 kV, Xd"= 0.2 pu.

Generator 3: 30 MVA, 20 kV, Xd"= 0.2 pu.

Transformer T1: 25 MVA, 220Y/13.8Δ kV, X=10%

Transformer T2: Single phase units, each rated 10 MVA, 127/18 kV, X=10%

Transformer T3: 35 MVA, 220Y/22 Y kV, X=10%

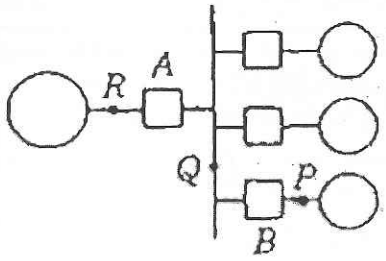


2. a. Discuss the importance of the short circuit study in power systems and discuss the classification of faults in power systems and compare them with respect to their severity. **03**

b. Explain what are Xd", Xd' and Xd of a synchronous machine. **04**

c. A 625 kVA generator with Xd" = 0.2 pu is connected to a bus through a circuit breaker as shown in fig. Connected through circuit breakers to the same bus there are three synchronous motors each rated 250 hp, 2.4 kV, unity pf with 90% efficiency with Xd" = 0.2 pu. The motors are operating at full load, unity pf and **07**

rated voltage with the load equally divided among the machines. Find the symmetrical short circuit current which must be interrupted through breakers A and B for a three phase fault at point P. Neglect the pre-fault conditions.



3. **a.** What are symmetrical components? How they are useful in solution of power system problems. **03**

b. Prove that a balanced set of three phase voltages or currents will have only positive sequence components. **04**

c. The sequence components of voltages across a three phase star connected impedance of (8+j6) Ohm/phase with a neutral impedance of j2 Ohm are $V_{a1} = 230\angle 0^0$, $V_{a2} = 60\angle 30^0$ and $V_{a0} = 40\angle -90^0$ volts. Determine the line currents and also the voltage drop across the neutral impedance. **07**

4. **a.** A three phase transposed transmission line has a self reactance of Xs pu and mutual reactance of Xm pu. Obtain the sequence impedances of the transmission line in terms of Xs and Xm. **06**

b. For the power system shown in figure 4b, obtain the negative sequence and zero sequence networks on a base of 30 MVA, 6.9 kV in the circuit of generator 1. **08**

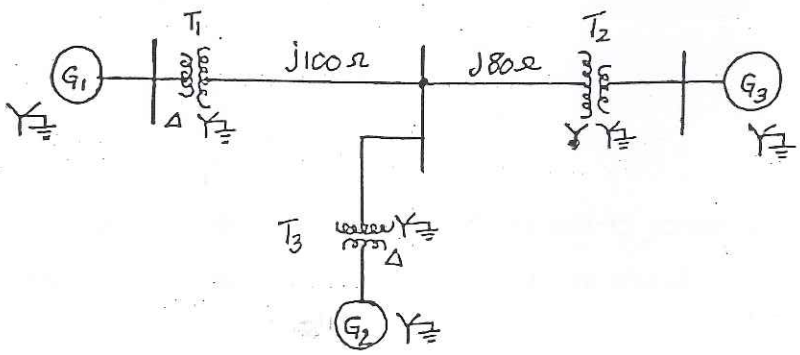


Fig.4b.

The details are::

Generator 1: 20 MVA, 6.9 kV, $X_1 = X_2 = 0.15$ pu, $X_0 = 0.10$ pu,

Generator 2: 10 MVA, 6.9 kV, $X_1 = X_2 = 0.15$ pu, $X_0 = 0.10$ pu,

Generator 3: 30 MVA, 13.8 kV, $X_1 = X_2 = 0.15$ pu, $X_0 = 0.10$ pu,

Transformer T1: 25 MVA, 6.9Δ – 115Y kV, with X = 10%,

Transformer T2: 30 MVA, 13Y– 130Y kV, with X = 10%,

Transformer T3: 12.5 MVA, 6.9Δ – 115Y kV, with X = 10%,.

The zero sequence reactance of the transmission lines are 300% of its negative sequence reactance value.

PART - B

5. **a.** Derive an expression for the fault current when a **line to ground fault** occurs at the terminals of a fully excited, unloaded three phase alternator. Show the inter connections of all the sequence networks to simulate this fault. **06**

b. A 20 MVA, 13.8 kV three phase alternator is having X1 = 0.2 pu, X2 = 0.35 pu and Xo = 0.10 pu is subjected to an L-L-G fault at its terminals. Find the fault current and the line-to-line voltages. Show the interconnection of the sequence networks. **08**

6. **a.** A 11 kV, 5 MVA generator is having X_1 = 0.3, X_2 = 0.2, Xo = 0.1 pu is feeding 4 MW at 11 kV, 0.8 pf to a 11 kV, 5 MVA synchronous motor having X_1 = 0.4, X_2 = 0.3, Xo = 0.1 pu. Both the machines are star connected and solidly grounded neutrals. Determine the fault currents when one line is open. **08**

b. An **LLG** fault occurs at the motor terminals in the power system shown in figure. Determine the fault current and the fault currents from the generator and the motor if the system were on no load and at a voltage of 6 kV at the fault point. **06**

The ratings are:

Generator: 50 MVA, 11.5 kV, X_1 = 0.3, X_2 = 0.2, Xo = 0.1 pu

Transformer T1: 45 MVA, 11/110 kV, X = 0.1 pu.

T2: 60 MVA, 115/6.6 kV, X = 10%.

Motor: 55 MVA, 6 kV, X_1 = 0.4, X_2 = 0.3 pu, Xo = 0.2 pu.

Transmission line: X_1 = X_2 = 45Ω, Xo= 200% of X_1.

Chose a base of 60 MVA, 110 kV in the transmission line.

7. **a.** Derive the power angle equation of a salient pole alternator connected to an infinite bus through an external reactance Xe and plot the power angle curve and **07**

draw the comments on it shape.

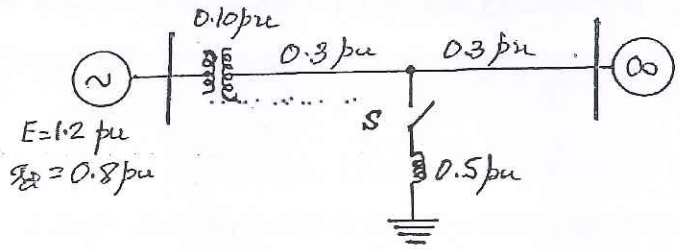
b. Derive the power angle equation of a salient pole alternator connected to an infinite bus through an external reactance Xe and plot the power angle curve and draw the comments on it shape. **07**

For the power system shown in fig. an inductor of reactance 0.5 pu is connected at the middle of the transmission line. The voltage of the infinite bus is 1.0 pu. Determine the steady state stability limit for the following conditions:

With the switch 'S' closed,

With the switch 'S' open,

With a capacitor of 0.5 pu reactance connected in series with the line with the switch 'S' open.



8. **a.** Derive the swing equation with usual notations. **04**

b. A generator is delivering 60% of maximum power to an infinite bus through a transmission line. A fault occurs such that the reactance between the generator and the infinite bus is increased to three times of its pre-fault value. When the fault is cleared the power that can be delivered is 80% of the original maximum value. Determine the critical clearing angle. **10**